

Economic assessment and comparison of acacia energy crop with annual traditional crops in Southern Europe

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ABSTRACT

In several policy documents bioenergy is recognized as an important renewable energy source in Italy. The increase in energy prices represents an opportunity for lignocellulosic energy crops such as acacia and poplar.

However, for Short Rotation Coppice (SRC) and Short Rotation Forestry (SRF) to be adopted by farmers, these crops must be perceived to be at least as profitable as crops that normally compete with these plantations for land use.

The purpose of this paper is to evaluate the economic feasibility of acacia (*Robinia pseudoacacia*) as an energy crop in a low input production regime in Italy and, in particular, to consider its competitiveness with wheat. Our results show that neither SRC and SRF techniques using assumed production costs (€3820 and €5285 ha⁻¹ yr⁻¹) nor biomass productions are able to obtain a positive profit (–€184 and –€172 ha⁻¹ yr⁻¹) that can convince farmers to invest in biomass plantations on their land. The results demonstrate that wheat is a more economically secure option than SRC or SRF.

The viability of local biomass production in Italy and Southern Europe depends on the active support of the governments; without them, biomass is not economically competitive for the farmers when compared to crops such as wheat.

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1. Introduction

Substituting biomass for fossil fuels in the generation of energy is an important strategy for the EU in order to mitigate climate change and enhance security of supply. For this purpose, bioenergy is being promoted through several EU Directives as well as through national policies in Southern Europe and specifically in Italy. Biomass-based electricity is being promoted in the Renewable Electricity Directive, which aims to increase the use of renewable energy sources (RES) to 22% by 2010 (European Parliament and Council, 2001).

In order to promote the renewable energy production in the specific study area, in 1999 the energy sector was liberalized in Italy making it the first country in Europe to adopt this resolution (Marinato Chiara, 2005) by means of Legislative Decree 77/99 (known as the Decreto Bersani) (Testo del, 2007). This, in turn, is derived from Electricity Directive 96/92/CE of the European

Parliament and Council, concerning common rules for the internal electricity market (Directive electricity).

Since its approval until 2006, Italy has achieved a renewable energy production of 14.6% (Marinato Chiara, 2005). Now a days in Italy, to encourage renewable energy production, new incentive-providing instruments have been approved such as the so-called Green Certificate (GC). Green Certificates, also known as Renewable Energy Certificates (RECs), are a tradable commodity, and they show that a certain amount of electricity is generated using renewable energy sources such as wind, solar, wave, tidal, geothermal hydro and biomass. Several countries favour the use of Green Certificates as a tool for supporting generation of green electricity that is closer to the market economy over more bureaucratic investment support and feed-in tariffs. Such national trading schemes are in use in Poland, Sweden, the UK, Italy, Belgium and in some states in the USA.

Italian GC forces the electricity sector (producers and importers) to reduce its non-renewable consumption sources and include an annual, targeted and increasing quota of renewable electricity. In 2002, for example, quotas of 2% were set and in 2008 the figure was 3.8% (Marinato Chiara, 2005).

In the specific case of agro forest biomass in Italy, the economic incentive applied in terms of GC considers that (a) power plants

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with outputs greater than 1 MWe can multiply their energy production expressed in MWh by 1.8 and in the case of (b) power plants with outputs less than 1 MWe can assume the same rule as those producing more than 1 MWe or assume a fixed tariff of €0.30 kWe⁻¹ (Testo del, 2007). In addition, GC tries to encourage the use of biomass produced at the local level by restricting the approval of GC only to those energy producers who obtain their biomass within a maximum radius of 70 km. The result of this policy is the creation of a market in which green energy can be sold with a greater profit margin in order to increase national renewable energy production and reduce the greenhouse gas emissions of the energy sector.

In fact, it would seem reasonable for Italy to expect new biomass production growth to generate clean energy over the next few years in order to achieve European energy goals expressed in the various directives. However, for the average farmer to adopt new crops such as Short Rotation Coppice or Forestry (SRC or SRF), the economics must favour these crops over traditional ones such as wheat, barley or maize, which usually compete for the same land.

According to the *Poplar white paper* (“Libro bianco della pioppicoltura”) (Gabriele, 2009) the surface area allocated to SRC and SRF for energy purposes was 5000 ha in 2008, basically located in the main 4 regions of the Po Plain (Piemonte, 300 ha, Lombardia, 3000 ha, Emilia Romagna 800 ha and Veneto, 800 ha); in the same area, the cultivation of cereals is widespread, and the surface area for wheat, the soil for which is seen as ideal for acacia stands, was 396,924 ha in the same period (ISTAT). Fig. 1 shows the crop distribution in the cultivatable soils of the Po Plain.

Although the current SRF and SRC surface area is minimal when compared with the main crops, in Italy there are more than 100,000 ha planted with agro forest species such as poplar, eucalyptus and acacia which demonstrate the high biomass production potential for energy purposes (Pettenella and Masiero, 2007).

Nevertheless the higher economic profitability is required as a sort of compensation for the higher perceived risk associated with growing SRC and SRF on the farms of the Po Plain, which have solid infrastructures and significant levels of mechanization; 69.5% of crop-growing surface in this area is represented by farms with a surface greater than 20 ha.

Hence, the primary purpose of this paper is to evaluate the economic feasibility of acacia (*Robinia pseudoacacia*) as an energy crop in a low input production regime in Italy and specifically to

consider its competitiveness with respect to conventional agricultural land uses.

The contribution that Green Certificates have made to the area of agricultural biomass has been estimated in order to determine whether this financial tool is efficient for developing a feasible energy system based on local biomass as an energy source. Wheat (*Triticum* sp.), which could compete for the same production areas, has been selected as the reference to compare with SRC and SRF.

1.1. Short rotation coppice and short rotation forestry

Even though both crops are technically “coppices” (i.e. stumps cut back to produce shoots), we will consider SRC as lignocellulosic energy crops cultivated in short rotation every two years over a total of 8 years and SRF is assumed to be a rotation every 5 years over 15 years (i.e. 3 cuttings), which are used to produce either heat or electricity or both (Venendaal et al., 1997). The plant species are: poplar (*Populus* spp.), willow (*Salix* spp.) and acacia (*Robinia pseudoacacia*), which have been tested and improved in Italy and in Europe in general for their biomass production potential by different research centres (Facciotto, 2003; Facciotto et al. 2007, 2006; Minotta, 2004). The interest in them as energy crops is based on (a) the possibility of obtaining higher biomass production than that from annual herbaceous crops and (b) their lower global environmental impact and energy consumption (Ericsson et al., 2006; Gasol et al., 2007, 2009; Heller et al., 2003).

2. Systems under study

Systems under study are acacia SRC and SRF and wheat. Wheat is used in this assessment as a comparison crop, as an alternative to farmer investment in areas with similar SRF demands and SRC acacia low input crops. The economic feasibility is determined by comparing the direct costs and prices of the two crops.

2.1. Agricultural reference system: wheat

The case study is located in the Piedmont region of Northern Italy where SRC or SRF as well as wheat can be planted. In these types of areas, the production of wheat is assumed to be 5 Mg grain ha⁻¹ with a sale price of €250 Mg⁻¹ (2008) and straw production is assumed to be 4 Mg ha⁻¹ with a market price of €52 Mg⁻¹. The average grain sale price between 1993 and 2008 was €160 Mg⁻¹ (Frascarelli, 2008). It is assumed that acacia and wheat enjoy the same area-based subsidy in the EU and so exclusion of this subsidy does not influence the viability of acacia in relation to wheat and is not considered in the study.

The average cost of wheat production in 2008 for this type of surface area is assumed to be €1150 ha⁻¹ (Guidi, 2008) taking into account increased diesel prices and the subsequent economic impact on the price of agrochemicals. The cost per Mg of grain in this study is taken to be €230 Mg⁻¹ (Guidi, 2008). Only production costs and income for 2008 and the period 1993–2007 have been considered for the economic assessment of wheat.

2.2. Acacia SRC and SRF: lignocellulosic energy crops under study

Previous studies on SRF and SRC in Northern Italy demonstrate that crops such as poplar (*Populus* spp.), cultivated in a high regime of agricultural inputs, generate prohibitive production costs causing the economic failure of the biomass production system from a farmer's point of view (Brun and Mosso, 2007).

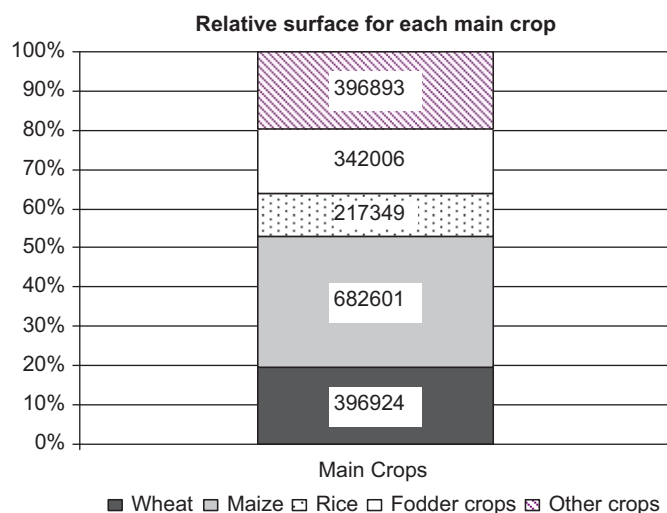


Fig. 1. Distribution of main crops in cultivatable soils of the Po Plain 2008–2009.

Table 1Labour sequence assumed in the biomass production for acacia (*Robinia pseudoacacia*) as an SRC (8 yr) and SRF (15 yr).

Operation			Tractor (A)	Implement (B)			A+B	Input rates
		Time (year)	Weight and power		Weight (kg)	Operating rate (h ha ⁻¹)	Fuel diesel (l ha ⁻¹ yr ⁻¹)	
Field preparation and crop Implantation	Ploughing (30–40 cm)	0	7.65 Mg. 176 kW	Plough	1590	1.60	44.70	–
	Disking the land	0	7.65 Mg. 176 kW	Harrow	980	0.5	14.00	–
	Acquiring the stems ¹	0	–	–	–	–	–	1100 plants ha ⁻¹
	Planting ¹	0	To rent					
Cycles	Line harrowing	SRC: 1,3,5 and 7	7.65 Mg. 176 kW	Harrow	980	1.20	20.00	× 7
		SRF: 1, 2, 6 and 11				1.20	20.00	× 9
Field recovery	Root killdown ¹	SRC: 8 SRF: 15	7.65 Mg 110 kW	To rent				

On the other hand, Acacia (*Robinia pseudoacacia*) is capable of growing without high inputs of agrochemicals, assuring remarkable biomass productions, and then the biomass production cost could be lower than of other SRFs and SRCs, e.g. poplar or willow. For these main reasons acacia is the crop selected in this assessment as a case study.

A Swedish 2-year rotation SRC model for a total 8-year cycle has been analysed. The SRF has been based on the American model with a 5-year rotation and a total cycle of 15 year. All of them have been cultivated with a low input technique that avoids agrochemical treatments and irrigation, and reduces the need for agricultural tending of plots, see Table 1.

The acacia dry biomass production assumed in this study was 7.5 Mg dry matter (0% water content) yr⁻¹ in the case of SRC and 9.60 dry matter (0% water content) yr⁻¹ in the case of SRF. This production is equivalent to 25 fresh Mg ha⁻¹ (40% w.c.) every two years in a SRC and 80 Mg ha⁻¹ (40% w.c.) produced every 5 years in the case of SRF. According to our experience and the bibliography consulted, these results are normal for acacia in the Piedmont region (Facciottio et al., 2008). Table 1 schematically shows agricultural labour performed in the acacia SRC and SRF.

Harvest and biomass transport are not present in the analysis because the sale price of biomass considered is a “stumpage price” (i.e. the price paid by the feller to the owner for the standing timber). In fact, such plantations are commonly sold as “standing” by the farmers who do not have machinery for harvesting or for chipping trees.

2.4. Methods

The economics of growing acacia (*Robinia pseudoacacia*) as an lignocellulosic energy crop is analysed at farm level by using a model presented by De Benedictis and Cosentino (1979) and applied previously in other studies (Brun and Mosso, 2007; Brun and Mosso) by means of the Excel model of professor Filippo Brun (Excel Database, 2008).

This model uses a total-step calculation method, in which all costs and revenues over the estimated lifespan of the plantation are discounted in order to assess the annual gross profit (according to the equation below):

$$\text{Annual gross margin} = \sum_{t=0}^T A_t(1+r)^{-t}$$

where r is the discount rate (3%), t the time (in years) at which a

Table 2

Machinery costs used to calculate the biomass production cost at farm level.

	Units	SRC	SRF
Tractor investment	€	83,872.00	
Plough cost	€	9677.00	
Harrow cost	€	3500.00	
Resale price of machinery and implements	€	9%–10% of the initial price	
Tractor and implements lifespan	yr	15	
Maintenance of machinery	€	0.05% € h ⁻¹ of the initial price	
Diesel price	€ l ⁻¹ (2008)	0.94	
Total diesel consumption without considering external contract labours	l ha ⁻¹	198.70	238.70
Lubricant consumption	l ha ⁻¹	3% of the diesel consumption	
Cost of lubricant	€ l ⁻¹	6.00	

payment (disbursement or revenue) is made or received, A_t the size of payment and T the total time period (in years).

Annual profits and the total cost incurred are calculated assuming that financial conditions are constant over the whole period. The sum of the annual gross margin, calculated as described, allows one to obtain the Net Present Value (NPV), which represents the economic result of an investment at present values. The economic results considered are: (A) production cost, given by the sum of all costs, including both internal and external costs; (B) net income, obtained by subtracting external costs from the gross marketable production (GMP) and (C) profit, calculated by subtracting all production costs from the GMP. This last indicator in our model is assumed as Net Present Value (NPV).

All the results shown in this paper (for all cultivations) are reported in surface area units of 1 ha.

2.4.1. Production cost

The cost of production includes (a.1) machinery costs including fuel and lubricant costs, garaging depreciation, maintenance and insurance costs, (a.2) specific services contracts and/or goods acquisition and (a.3) general external costs such as administration costs, etc. Internal costs are also considered, e.g. (b.1) labour

payments, (b.2) land remunerations and (b.3) working capital remunerations (interest).

In order to calculate the machinery cost, machinery replacement and machinery maintenance cost, the data shown in Table 2 have been assumed.

2.4.2. Service contract and goods acquisition

The service contracts assumed for SRC and SRF are crop planting, with an assumed cost of €400 ha⁻¹ and the root killdown, assuming €700 ha⁻¹ in accordance with pricing for the sector (Personal Communication, 2008a). A price of 0.5 Euros per stem, including transport, has also been assumed (Personal Communication, 2008a).

2.4.3. Labour remuneration

The total working hours necessary for the SRC is 10.5 and 16 h ha⁻¹ for the SRF contracted operations are not included in these values because they are considered external services. A remuneration of 11 Euros per hour has been assumed for all internal operations where the labour is carried out by the farmer.

2.4.4. Land remuneration

Considering the characteristics of the land ordinarily used for these kinds of crops, land has been assumed to have a value of about €20,000 ha⁻¹. Therefore, by adopting an interest rate of 1% one obtains an annual land use cost of €200 ha⁻¹ yr⁻¹.

Table 3

Comparison of economic results of acacia biomass plantations as SRC in 8-year cycles and SRF in 15-year cycles.

	SRC			SRF		
	Total (€ ha ⁻¹)	Annual amount (€ ha ⁻¹ yr ⁻¹)	%	Total (€ ha ⁻¹)	Annual amount (€ ha ⁻¹ yr ⁻¹)	%
Stems (1)	410.00	34.31	10.73	410.00	34.31	7.76
Plantation (2)	400.00	33.51	10.47	400.00	33.51	7.57
Roots cutting (3)	552.59	46.29	14.47	449.00	37.64	8.49
Total service and goods acquisition (a)=1+2+3	1362.19	114.11	35.67	1259.00	105.46	23.82
Machinery cost (4)	148.28	12.42	3.88	196.00	16.45	3.71
Diesel and lubricants (5)	208.07	17.43	5.45	300.00	25.15	5.67
Total cost of machinery (b)=4+5	356.35	29.85	9.33	497.00	41.60	9.39
General external cost (c)	561.94	47.07	14.70	956.00	80.05	18.09
Total external cost (d)=a+b+c	2280.48	191.03	59.70	2712.00	227.11	51.30
Labour remuneration (6)	107.83	9.03	2.82	149.00	12.52	2.82
Land remuneration (7)	1403.94	117.60	36.75	2388.00	200.00	45.18
Working capital remuneration (8)	27.83	2.33	0.73	37.00	3.12	0.70
Total internal cost (e)=6+7+8	1539.60	128.97	40.30	2574.00	215.63	48.70
Total cost of the process € ha ⁻¹ (f)=d+e	3820	320	100	5286	442.74	100
Production cost of the biomass (40% w.c. standing trees)	38.20			22.00		

Table 4

Synthesis of the economic results updated for acacia and wheat ($r=0.03$).

	Wheat 2008	Wheat 1993–2007	SRC		SRF	
	€ ha ⁻¹ yr ⁻¹		€ ha ⁻¹	€ ha ⁻¹ yr ⁻¹	€ ha ⁻¹	€ ha ⁻¹ yr ⁻¹
Gross marketable production (GMP)	1250+208 ^a =1458	725+208=933 ^b	1297.00	108.62	2698	226.03
Energy bonus			316.00	26.46	537.00	45.00
Production cost=(EC+IC)	1150 ^c	850 ^d	3820.00	319.99	5285.00	442.74
External cost (EC)			2280.00	191.02	2711.00	227.11
Internal cost (IC)			1540.00	128.97	2574.00	215.63
Net income			-668.00	-55.94	524.00	43.92
Profit or VAN	308	83	-2207.00	-184.91	-2050	-171.70

^a Considering €208 of straw sale for 2008 in Italy and grain sale price of €250 Mg⁻¹.

^b Considering €208 of straw sale and the average grain sale price (€160 Mg⁻¹) from 1993 to 2007.

^c Production cost in 2008 [15].

^d Production cost in 2007 [15].

2.4.5. Working capital remuneration

Working capital to be taken into account is related to machinery and implements. The cost has been estimated as the investment value multiplied by a real (i.e. without inflation) rate of 3%.

2.4.6. Biomass sale price

The sale price of biomass assumed in this study is €15 Mg⁻¹ of standing biomass (40% water content) in accordance with prices currently in use in Northern Italy under similar conditions. Another reference price in the Italian biomass market is the sale price of the chipped biomass, which is taken as €42–47 Mg⁻¹ (40% w.c.) (Personal Communication, 2008b).

2.5. Characteristics of data used

The experimental data related to costs and incomes have been obtained from real farms located in the Piedmont and Lombardy plains, using current (2008) prices. In order to compare a larger amount of information, interviews with expert agronomists have also been carried out (Brun and Mosso).

3. Results and discussion

All the results obtained and shown in Table 3 referring to the SRC and SRF production costs have been updated with the planting time as reference (year=0).

As per assessment of total cost of the process, the SRF (15 years) technique is more costly than the SRC (8 years), giving a total cost of €5286 ha⁻¹ compared with €3820 of the SRC, due to the longer cycle and the consequent influence of the land cost. However, because biomass production is higher (58.3%) in the SRF, its unitary production cost, expressed in €Mg⁻¹, turns out to be significantly lower (22.00) than the SRC (38.20).

According to our results, total external cost (sum of the costs *a*, *b* and *c*, see Table 3) for acacia low input production contributes most to total production cost (59.70% in the case of SRC and 51.30% in the case of SRF). Furthermore, these kinds of expenses are incurred mainly during the establishment of the plantation. These affect the farmers' liquid assets since they are incurred during the first and second years of the lifespan of the plantation. The internal cost contributes to the remaining 40.30% and 48.70% for SRF and SRC, respectively.

Other economic indicators, reported in Table 4, show that the profit for biomass production is always negative and SRC fares worst as energy crop. For SRC the Gross Marketable Production (GMP) does not cover the external cost hence, both net income and final profits are negative. In the case of SRF, GMP is enough to cover external costs hence, the net income is positive although final profit remains negative, see Table 4.

The economic balance described in Table 4 demonstrates that both SRC and SRF techniques, with assumed production costs and biomass production, are undoubtedly unable to produce a positive profit and thus unable to convince farmers to invest in biomass plantations.

When acacia biomass production is compared with wheat production, it is evident that wheat is an economically more secure option for the farmer than SRC or SRF. The profit for wheat is always positive not only with current high prices but also considering the average of the last 14 years.

One of the main reasons for the failure of acacia SRC and SRF lies in the value of the product, €15 Mg⁻¹ of standing biomass (40% water content). Other SRF and SRC crops consulted in the bibliography present a biomass sale price of €63 Mg⁻¹ of chipped biomass and even €70, €100 and €130 Mg⁻¹ of chipped biomass in comparison with €42–47 Mg⁻¹ (40% w.c.). In these cases biomass systems seem to be economically feasible from the farmer's point of view but these gross margins are based on peak-productivity combustible yields (14 and 10 Mg 20% w.c.) (Styles et al., 2008).

4. Sensitivity analysis

Sensitivity analysis allows the study of the impact of varying inputs in a model in which uncertainty can affect the results. In more general terms, it investigates the robustness of the results, giving information about how much an objective can change.

In order to investigate how biomass production and sale price, which are fixed in our model, influence the economics of growing acacia as an SRC or SRF, we have simulated their variation within a reasonable range. The results can be seen in Figs. 2 and 3.

Fig. 2 illustrates the point with the default values used in the model taking an average production of 7.50 Mg ha⁻¹ yr⁻¹ and a sale price of €15.00 Mg of biomass (40% w.c.), values which produce a negative profit in the –200 to –100 range.

According to our assessment, in order to obtain a positive profit for biomass productions of 10 Mg ha⁻¹ yr⁻¹, which could be achieved with agronomic improvements, the price of biomass would need to be approximately €30–32 fresh Mg⁻¹. As an alternative, with current production of 7.50, a positive profit can only be achieved with prices higher than €38 fresh Mg⁻¹ (outside the range under consideration here).

Similarly, Fig. 3 shows the simulations carried out for SRF.

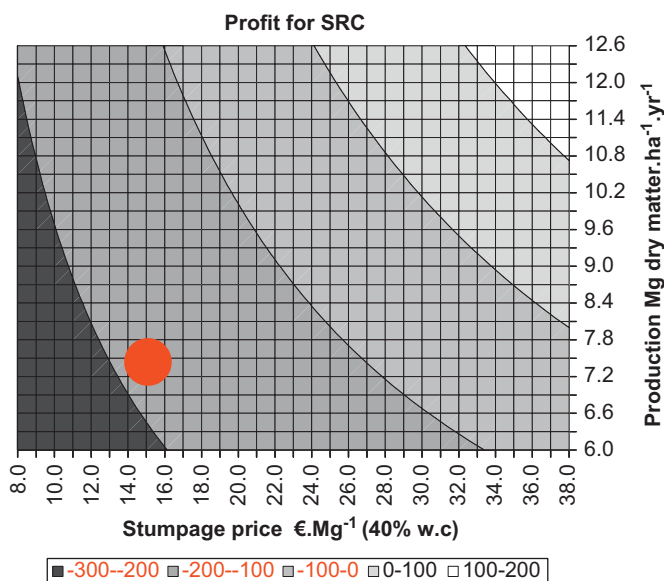


Fig. 2. Feasibility (Profit) of acacia SRC with biomass production and sale price variations.

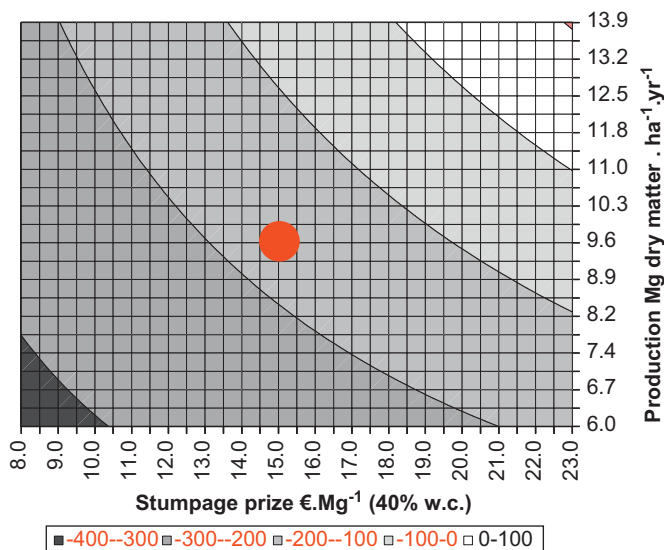


Fig. 3. Feasibility of acacia SRF with biomass production and sale price variation.

Fig. 3 demonstrates the point considered for the main estimations with an average production of 9.50 Mg ha⁻¹ yr⁻¹ and a sale price of €15.00 Mg. These values produce a negative profit included within the –200 to –100 range.

In this case the break even point is reached only in the upper right-hand area, with production greater than 11.8 and prices higher than €19. For production of 9.50, a positive profit can be achieved at a price of (around) 24 Euros.

4. Conclusions

Wood is a renewable source of energy, which has a high potential for covering an increase in energy demand as a result of its environmental advantages. Despite these advantages (such as carbon neutrality), bioenergy produced by energy crops has not evidenced widespread interest among the general public, let alone among

private companies. To some extent, this fact can be attributed to the high production cost and low biomass sale price, which cause, together, the economic failure of energy crops production systems.

However acacia SRF with 5 cuttings presents slightly better results than SRC with a biennial rotation neither are competitive crops in terms of plantation in Northern Italian regions, under current market conditions and sale prices of biomass, from a farmer's point of view.

The most competitive land use clearly favours traditional annual crops such as wheat. In addition, the biomass production considered for acacia SRC and SRF in the case study does not generate sufficient Gross Marketable Production (GMP) under the current sales price of biomass in Italy (€15 fresh Mg⁻¹) to produce a positive profit or even a net income in the case of SRC.

While energy generation plants pay approximately €70 for 1 Mg of fresh biomass (40% w.c.), the farmers only receive €15 for standing biomass or €45 for chipped biomass. The result is the same in both cases: insufficient GMP to produce a positive profit.

Wheat grain sales price worth 10 times the cost of chip wood cultivation and production cannot be suppressed to less than a certain level, not only because there are high external costs in acacia SRF and SRC but also because internal costs (as labour or land rent) are very high.

Simulations demonstrate that even in the case where both biomass cultivation techniques can obtain higher production levels than the existing one, the sale price of biomass would have to be higher than the current one in order to achieve positive results. The optimization of agricultural labour and the biomass production obtained can help to increase the feasibility of the bioenergy systems, but they are not the key factors in assuring the implementation of biomass at a local scale of production. As long as SRC and SRF biomass are considered low-quality wood products, they will be infeasible from the farmer's point of view.

Both public funding and the Green Certificate programme will need to reduce the production cost or increase the sales price of biomass in Italy if biomass is to be used as a renewable source for energy generation. To illustrate this, in order to achieve the break even point of €24 Mg⁻¹ (stumpage price) for SRF or over €38 for SRC, energy producers would have to transfer 5% of the Green Certificate's value to the farmers in the case of SRF and 13% in the case of SRC.

These amounts seem reasonable even if market operators to date have made different choices by using imported biomass as a structural resource. In the long term, this strategy appears to us unsustainable.

Hence, the major question is how to value the purported benefits associated with biomass, such as a decrease in greenhouse gas emissions or increased security of energy supply. If these obvious benefits are left unvalued, biomass may not be as competitive with other energy sources as anticipated. Furthermore, the costs of subsidizing the price difference between biomass and fossil fuels can be viewed in terms of CO₂ emissions saved, which is not purely an economical issue as it is a political one also. This fact complicates government actions such as allocating subsidies and tax benefits.

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